ECON 5140: Applied Econometrics

Homework 3 — Solutions: Problems 1–4

Problem 1: Potential Outcomes and Causal Estimands

Data

I	Y(1)	Y(0)	D	Y_OBS	T_I = Y(1)−Y(0)
1	45	35	1	45	**+10**
2	50	42	1	50	**+8**
3	38	40	0	40	**−2**
4	55	48	1	55	**+7**
5	42	38	0	38	**+4**
6	48	45	1	48	**+3**
7	40	44	0	44	**−4**
8	52	46	0	46	**+6**

Part (a): ATE, ATT, ATU

ATE = E[Y(1) − Y(0)] — average over all 8 units:

$$\text{ATE} = \frac{10 + 8 + (-2) + 7 + 4 + 3 + (-4) + 6}{8} = \frac{32}{8} = \boxed{4} \tag{1}$$

ATT = E[τ_i | D=1] — average over treated units {1, 2, 4, 6}:

$$\text{ATT} = \frac{10 + 8 + 7 + 3}{4} = \frac{28}{4} = \boxed{7} \tag{2}$$

ATU = E[τ_i | D=0] — average over control units {3, 5, 7, 8}:

$$\text{ATU} = \frac{(-2) + 4 + (-4) + 6}{4} = \frac{4}{4} = \boxed{1} \tag{3}$$

Part (b): Simple Difference in Outcomes (SDO)

$$E[Y \mid D = 1] = \frac{45 + 50 + 55 + 48}{4} = \frac{198}{4} = 49.5 \tag{4}$$

$$E[Y \mid D = 0] = \frac{40 + 38 + 44 + 46}{4} = \frac{168}{4} = 42.0 \tag{5}$$

$$\text{SDO} = 49.5 - 42.0 = \boxed{7.5} \tag{6}$$

SDO ≠ ATE (4 ≠ 7.5). The reason is that treatment is not randomly assigned. Units who received training (D=1) differ systematically from those who did not—they have higher baseline potential outcomes Y(0) and larger individual treatment effects. This introduces **selection bias** and **heterogeneous treatment effect (HTE) bias**, causing SDO to overstate the true average causal effect.

Part (c): Decomposition of SDO

The exact decomposition is:

$$\text{SDO} = \underbrace{\text{ATE}} + \underbrace{(E[Y(0) \mid D = 1] - E[Y(0) \mid D = 0])}_{\text{Selection Bias}} + \underbrace{p \cdot (\text{ATT} - \text{ATU})}_{\text{HTE Bias}} \tag{7}$$

Step 1 — Selection Bias:

$$E[Y(0) \mid D = 1] = \frac{35 + 42 + 48 + 45}{4} = \frac{170}{4} = 42.5 \tag{8}$$

$$E[Y(0) \mid D = 0] = \frac{40 + 38 + 44 + 46}{4} = \frac{168}{4} = 42.0 \tag{9}$$

$$\text{Selection Bias} = 42.5 - 42.0 = \boxed{0.5} \tag{10}$$

Step 2 — HTE Bias:

$$p = \frac{4}{8} = 0.5, \quad \text{ATT} - \text{ATU} = 7 - 1 = 6 \tag{11}$$

$$\text{HTE Bias} = 0.5 \times 6 = \boxed{3} \tag{12}$$

Verification:

$$\text{ATE} + \text{Selection Bias} + \text{HTE Bias} = 4 + 0.5 + 3 = 7.5 = \text{SDO}\checkmark \tag{13}$$

Part (d): Effect of Random Assignment

Under random assignment, D is independent of the potential outcomes: D ⊥ {Y(0), Y(1)}.

- **Selection bias → 0:** E[Y(0)|D=1] = E[Y(0)|D=0], so there is no systematic baseline difference between groups.
- **HTE bias → 0:** ATT = ATU = ATE, because the distribution of treatment effects is the same in the treated and control groups.

Therefore, **SDO → ATE**, and the naive comparison of means becomes a valid causal estimator. This is the fundamental justification for randomized controlled trials.

Problem 2: DAGs — Confounding and Colliders

Part (a): Confounder

DAG: D ← X → Y, D → Y

X is a confounder (common cause of both D and Y). It opens a *backdoor path*: D ← X → Y.

Why naive comparison is biased: A simple comparison of outcomes by treatment status conflates the causal effect D → Y with the spurious association flowing through the backdoor path. Units with high X are both more likely to be treated and have higher Y, so the treatment-control difference in observed outcomes reflects X's influence, not just D's.

Adjustment: Block the backdoor path by conditioning on X — e.g., include X as a covariate in OLS regression, match on X, or stratify by X. By the backdoor criterion, controlling for X is both necessary and sufficient to identify the causal effect of D on Y.

Part (b): Collider

DAG: D → X ← Y

Are D and Y independent in the population?
Yes. There is no open path between D and Y without passing through X. Since X is a collider (both arrows point *into* it), the path D → X ← Y is **blocked** by default. D and Y are marginally independent.

What happens when we condition on X?
Conditioning on a collider *opens* the previously blocked path, inducing a spurious statistical association between D and Y. This is known as **collider bias** (or Berkson's paradox). Even though D has no causal effect on Y, conditioning on X makes them appear correlated — creating a misleading conclusion.

Real-world example:
Consider studying the relationship between athletic ability (D) and academic performance (Y) among students *admitted to a selective university* (X). Admission is a collider — both athletic ability and academic performance increase the probability of admission. In the general population, the two traits may be unrelated. But within admitted students, athletes may appear to have lower academic performance (and vice versa), creating a spurious negative correlation that doesn't exist in the population. Conditioning on admission (collider) opens the backdoor and induces this bias.

Problem 3: Exact vs Approximate Matching

Part (a): Exact Matching

Definition: Exact matching pairs each treated unit with a control unit that has *identical* values on every observed covariate. The matched comparison then estimates ATT by averaging treatment-control outcome differences within each matched stratum.

When it works well:

- Covariates are discrete or categorical (few possible values)
- Covariate space is small (few covariates)
- Dataset is large enough that many exact matches exist

Curse of Dimensionality: As the number of covariates k grows, the covariate space expands exponentially. Even with many observations, the probability that any two units share the same value on all k covariates shrinks rapidly. In practice, most treated units remain unmatched, leading to a small, possibly unrepresentative matched sample and severely reduced effective sample size. Exact matching becomes computationally and statistically infeasible in high dimensions.

Part (b): Approximate (Distance-Based) Matching

Definition: Rather than requiring identical covariate values, approximate matching pairs each treated unit with the control unit(s) closest to it according to some distance metric, tolerating small discrepancies ("approximate" matches).

Two common distance metrics:

1. **Mahalanobis Distance:**

$$d(x_i, x_j) = \sqrt{(x_i - x_j)^T \Sigma^{-1} (x_i - x_j)} \tag{14}$$

Accounts for covariate scale and correlations by normalizing using the covariance matrix Σ.

2. **Propensity Score Distance:**

$$d(x_i, x_j) = |e(x_i) - e(x_j)| \tag{15}$$

Collapses all covariates into a single scalar—the estimated probability of treatment—and matches on that. By Rosenbaum & Rubin's theorem, matching on the propensity score balances all covariates in expectation.

How Nearest-Neighbor (NN) Matching Works:
For each treated unit i, find the control unit j* that minimizes the distance d(x_i, x_j) over all control units. This forms a matched pair (i, j*). The ATT estimate is the average difference in outcomes across all matched pairs:

$$\widehat{\mathrm{ATT}}_{\mathrm{NN}} = \frac{1}{n_1} \sum_{i:D_i=1} \left(Y_i - Y_{j^*(i)}\right) \tag{16}$$

Practical choices include: matching with or without replacement, 1:1 vs 1:k matching, and imposing a *caliper* (maximum allowable distance) to discard treated units without a sufficiently close match.

Problem 4: Inverse Probability Weighting (IPW) — Numerical Example

Data

I	D	Y	E(X)
1	1	52	0.60
2	1	48	0.80
3	0	42	0.30
4	0	38	0.20
5	1	55	0.70
6	0	45	0.40

Part (a): ATE Weights

$$w_i^{\text{ATE}} = \frac{D_i}{e(X_i)} + \frac{1 - D_i}{1 - e(X_i)} \tag{17}$$

I	D	Y	E(X)	CALCULATION	W_ATE
1	1	52	0.60	1 / 0.60	**1.6667**
2	1	48	0.80	1 / 0.80	**1.2500**
3	0	42	0.30	1 / (1−0.30) = 1/0.70	**1.4286**
4	0	38	0.20	1 / (1−0.20) = 1/0.80	**1.2500**
5	1	55	0.70	1 / 0.70	**1.4286**
6	0	45	0.40	1 / (1−0.40) = 1/0.60	**1.6667**

Part (b): ATT Weights

$$w_i^{\text{ATT}} = D_i + (1 - D_i) \cdot \frac{e(X_i)}{1 - e(X_i)} \tag{18}$$

Treated units always receive weight 1; control units receive weight e(X)/(1−e(X)).

I	D	Y	E(X)	CALCULATION	W_ATT
1	1	52	0.60	1 (treated)	**1.0000**
2	1	48	0.80	1 (treated)	**1.0000**
3	0	42	0.30	0.30 / 0.70	**0.4286**
4	0	38	0.20	0.20 / 0.80	**0.2500**
5	1	55	0.70	1 (treated)	**1.0000**
6	0	45	0.40	0.40 / 0.60	**0.6667**

Part (c): IPW Estimate of ATE

Using the Horvitz-Thompson IPW estimator:

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{D_i Y_i}{e(X_i)} - \frac{(1-D_i)Y_i}{1-e(X_i)}\right] \tag{19}$$

Treated contribution (subtract control contribution):

I	D	Y	E(X)	D·Y/E(X)	(1−D)·Y/(1−E)
1	1	52	0.60	86.667	0
2	1	48	0.80	60.000	0
3	0	42	0.30	0	60.000
4	0	38	0.20	0	47.500
5	1	55	0.70	78.571	0
6	0	45	0.40	0	75.000
Σ				**225.238**	**182.500**

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{1}{6}(225.238 - 182.500) = \frac{42.738}{6} \approx \boxed{7.12} \tag{20}$$

Part (d): IPW Estimate of ATT

$$\widehat{\mathrm{ATT}}_{\mathrm{IPW}} = \frac{1}{n_1} \sum_{D_i=1} Y_i \; - \; \frac{\sum_{D_i=0} w_i^{\mathrm{ATT}} \cdot Y_i}{\sum_{D_i=0} w_i^{\mathrm{ATT}}} \tag{21}$$

Treated mean:

$$\bar{Y}_1 = \frac{52 + 48 + 55}{3} = \frac{155}{3} \approx 51.667 \tag{22}$$

Weighted control mean:

Numerator:

$$0.4286 \times 42 + 0.2500 \times 38 + 0.6667 \times 45 = 18.000 + 9.500 + 30.000 = 57.500 \quad (2$$

Denominator (sum of control weights):

$$0.4286 + 0.2500 + 0.6667 = \frac{3}{7} + \frac{1}{4} + \frac{2}{3} = \frac{36 + 21 + 56}{84} = \frac{113}{84} \approx 1.3452 \quad (24)$$

Weighted control mean:

$$\bar{Y}_0^{\,w} = \frac{57.500}{1.3452} \approx 42.748 \tag{25}$$

$$\widehat{\mathrm{ATT}}_{\mathrm{IPW}} = 51.667 - 42.748 \approx \boxed{8.92} \tag{26}$$

Part (e): Largest ATE Weight — Interpretation

Units 1 and 6 both have the largest ATE weight of **1.6667** (tied).

- **Unit 1** (D=1, e=0.60): A treated unit with a relatively low propensity score. Among all treated units (e = 0.60, 0.80, 0.70), unit 1 had the lowest probability of receiving treatment. Being treated despite a lower propensity makes it "surprising," so it receives a larger upward weight on the treated side.
- **Unit 6** (D=0, e=0.40): A control unit with the highest propensity score among controls (e = 0.30, 0.20, 0.40). Having a higher chance of treatment but ending up in the control group makes it "surprising," so it receives a larger upward weight

on the control side.

General principle: IPW assigns larger weights to units whose observed treatment status was *unlikely* given their covariates. This reweights the sample so that treated and control groups become comparable in covariate distribution — units near the margin of treatment carry more information for identifying the causal effect.